

- Q.32 Discuss the manufacturing process of butanol, including the synthesis routes, catalyst selection, and process description.
- Q.33 Provide a detailed flowsheet illustrating the key steps involved in the production of ethanol, from feedstock preparation to product purification.
- Q.34 Difference between aromatic compound over c3 compound.
- Q.35 Identify major engineering problems encountered in the manufacturing process of C1 compound (methanol), such as catalyst deactivation, energy consumption, and environmental consideration.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe the manufacturing process of phenol, detailing the synthesis methods, raw material selection, and process optimization techniques.
- Q.37 Explain the properties of C3 compounds, with a focus on acetone, including its molecular structure, physical properties, and industrial uses.
- Q.38 Write a short note on any three:-
- 1 Butanol
 - 2 Petrochemical
 - 3 Petroleum Product
 - 4 IUPAC naming

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180651

Roll No.

5th Sem / P & P

Subject:- Petroleum Refining and Petrochemicals

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which method is commonly used to transport crude oil?
- a) Pipeline
 - b) Railway
 - c) Tanker ships
 - d) All of the above
- Q.2 What is the boiling range of diesel fuel?
- a) 40-70°C
 - b) 70-180°C
 - c) 180-260°C
 - d) 260-350°C
- Q.3 What is the IUPAC name for CH₃-CH₂-CHO?
- a) Propanal
 - b) Butanal
 - c) Pentanal
 - d) Hexanal
- Q.4 What is the primary use of Methanol?
- a) Fuel additive
 - b) Solvent
 - c) Feedstock for plastics
 - d) All of the above
- Q.5 Which process is commonly used for the manufacturing of Ethylene?
- a) Steam cracking
 - b) Catalytic reforming
 - c) Hydrogenation
 - d) Alkylation
- Q.6 Which compound is commonly used as a precursor for the synthesis of nylon?
- a) Phenol
 - b) Benzene
 - c) Toluene
 - d) Xylene

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- Q.7 Which C3 compound is commonly used in nailpolish removers?
 a) Acetone b) Methanol
 c) Ethanol d) Propanol
- Q.8 Petrochemical manufacturing units differ from commodity chemical plants.
 a) True b) False
- Q.9 Sweetening of petroleum product means the removal of
 a) Wax
 b) Water
 c) organic impurities
 d) Sulphur & its compounds
- Q.10 Production of plastics like PVC is manufacture by the use of _____
 a) C4 compound b) C1 compound
 c) C3 compound d) C2 compound

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 LPG is _____
- Q.12 Write one use of c1 compound?
- Q.13 The chemical composition of carbon in petroleum is _____
- Q.14 Petroleum is a complex mixture of hydrocarbons that occurs in the sedimentary rocks _____
- Q.15 Name any one petrochemical plant in India.
- Q.16 Write formula of specific gravity.
- Q.17 Write one property of vinyl chloride?
- Q.18 Write IUPAC name of aldehyde.

- Q.19 Write one application of c2 compound?
- Q.20 Define Petroleum

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 List major petrochemical plant in India.
- Q.22 Write about properties & uses of vinyl chloride
- Q.23 Explain the purification techniques employed in refining petroleum products to meet quality standards and regulatory requirements.
- Q.24 Describe about the methanol with flowsheet.
- Q.25 Explain the principles of IUPAC nomenclature for naming cyclic compounds, and aromatic compounds, emphasizing systematic rules for naming organic molecules.
- Q.26 Identify the raw materials used in petrochemical production, focusing on petroleum-derived feedstocks and their availability and sourcing.
- Q.27 Present a flow sheet of a typical petroleum refinery, illustrating the key processes involved in refining crude oil into valuable products.
- Q.28 Explain the manufacturing processes for the production of ethanol.
- Q.29 Discuss transportation of crude oil and its product.
- Q.30 Describe about the composition and classification of petroleum, highlighting the primary hydrocarbon components and other impurities.
- Q.31 Outline the manufacturing process of acetone, raw material and process optimizing strategies.